Fundamental Information-Theoretic Limits of Behavioral AI Agent Detection in Digital Markets

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Abstract

Autonomous software agents now participate in digital markets, elections, and public discourse, prompting regulators on three continents to draft behavioral-detection regimes. We ask whether such regimes can, in principle, identify AI agents from observable behavior alone. Using Fano's inequality, we prove that any classifier that maps behavioral features X to agent category Y obeys an accuracy ceiling determined by the mutual information $I(X;Y)$ and the label entropy $H(Y)$. Applied to an 8-category taxonomy spanning 2,744 blockchain addresses—a setting chosen because its public transaction record allows exact measurement of $I(X;Y)$—we find a ceiling of 97.2%; the best supervised classifier reaches 95.3%. The gap is 1.7–4.6 percentage points across feature sets: behavioral detection is already near the information-theoretic limit. Unsupervised discovery saturates at 2–3 clusters regardless of feature set (HDBSCAN: 2; K-Means silhouette: 3). A mechanistic decomposition separates two signal channels: execution speed is a *tool* signature (LLM-driven DeFi operations run at 216 s vs. 289 s for rule-based agents) while inter-decision interval is an *agent* signature (general reaction time: 441 s vs. 321 s; Cohen's $d = 1.01$). These findings imply that purely behavioral regulation cannot reliably identify AI agents across contexts; durable governance requires either provenance registration or execution-layer signatures.

Keywords: artificial intelligence; agent detection; information theory; Fano's inequality; AI governance; digital markets.

Introduction

Autonomous software agents—programs that perceive a digital environment, decide without per-action human approval, and execute actions that move money, vote, or speak—have left the laboratory. They trade on derivatives venues, respond to constituents on social platforms, and settle governance outcomes in distributed ledgers. Market analysts estimate that tens of billions of U.S. dollars in annual transaction value on public blockchains alone now flow through such agents[1, 2]; comparable (though harder to audit) activity occurs on advertising, e-commerce and social platforms. The European Union's AI Act[3], the United States' Executive Order on AI safety[4], and the United Kingdom's pro-innovation AI regulation White Paper[5] all require that "high-risk" deployments be identified, disclosed, or certified. Each proposal presupposes

that AI agents can be reliably distinguished from humans, from simple automation, and from one another.

The natural detection substrate is *behavior*: an agent is known by its actions in a public channel. Platforms that mandate real-name provenance (such as certified identity registries) are legally and technically fragile in the presence of pseudonymous participation, off-shore hosting, and cross-border deployment. Behavior, by contrast, is visible everywhere a platform offers an API. Almost every recent detection proposal—whether for social-media bots[6, 7], frontrunning agents[1, 8], coordinated inauthentic behavior[9], or AI-generated content[10, 11]—therefore relies on statistical classifiers trained on behavioral features. The unstated assumption is that classifier accuracy can, in principle, be driven arbitrarily close to 100% by collecting more features, labeled data, or model capacity.

The stakes of this assumption are not abstract. The rapid proliferation of agentic foundation-model systems[12, 13, 14, 15] has focused policy-maker attention on three empirical questions. First, *how many* AI agents are now active in a given platform—a prevalence question. Second, *which kind* of AI agent is active at a given moment of activity—a categorical question that underlies any category-specific disclosure or risk assessment. Third, *when* an agent is acting on behalf of a human versus pursuing emergent objectives—a question that underlies any accountability attribution. Every one of these questions is typically answered, in practice, by training a behavioral classifier; and every one is governed by the same information-theoretic ceiling. Our analysis applies most directly to the categorical question, but the methodology—estimate $I(X;Y)$, invert Fano, report the head-room—generalizes to the others.

We show that this assumption is wrong, and quantify how wrong it is. Treating the classification pipeline $X \to \hat{Y}$ as a communication channel, Fano's inequality[16] supplies an accuracy ceiling that depends only on the label entropy $H(Y)$ and the mutual information $I(X;Y)$ between features and labels. We estimate both quantities directly on a corpus of 2,744 real agent deployments, using a setting in which the complete action history is publicly observable: transactions on the Ethereum public ledger. On this corpus, across three increasingly expressive behavioral feature sets (X_{23}, X_{31}, X_{47}), the tightest achievable ceiling is 97.2%. Real classifiers trained on the same data reach 95.3%. *The gap is roughly two percentage points: behavioral detection is already at the information-theoretic limit.*

Crucially, this limit is not a consequence of dataset size, labeling noise, or specific model choice. It is a property of the observation channel itself. Adding features increases $I(X;Y)$ only up to a saturation point set by the confusability of behavioral categories; beyond that point, further engineering yields vanishing returns. We show this empirically: going from 23 to 47 features raises the ceiling by less than one percentage point.

Three findings follow. First, a *mechanistic* decomposition shows that the residual signal lives almost entirely in inter-decision *timing*, not in transactional content. LLM-driven agents execute individual decentralized-finance (DeFi) operations *faster* than rule-based managers (216 s vs. 289 s median)—because the tool-side code path is identical—but their general inter-transaction interval is *slower* (441 s vs. 321 s), reflecting language-model inference latency. Execution speed is thus a *tool* signature; decision interval is an *agent* signature. Second, unsupervised discovery collapses: both K-Means and density-based clustering (HDBSCAN) saturate at 2–3 natural

behavioral clusters regardless of feature-set richness, even as the taxonomy itself distinguishes eight. Third, the ceiling generalizes across architectures: a replication on the Polygon proof-of-stake chain, with different gas economics, yields a cross-chain transfer accuracy of only 22.2%, consistent with an information-theoretic rather than engineering bound.

The policy consequence is immediate. Behavioral regulation of AI agents is structurally limited to coarse tiers (roughly: static automation, rule-driven automation, adaptive agents). Finer-grained requirements—for instance, stricter disclosure obligations specifically on LLM-driven agents—*cannot* be robustly enforced through behavioral observation. Jurisdictions that require such distinctions will have to either (a) mandate provenance registration at the point of deployment, or (b) compel disclosure of execution-layer signatures (for example, cryptographic attestations of the inference runtime). Our results give the first principled, quantitative basis for this trade-off.

The choice of blockchain as a measurement substrate deserves brief justification. The argument is domain-general, but the quantitative claim requires that we *know* the ground-truth categories for a large and diverse sample of deployed agents, and that we observe their full action history. On closed platforms (advertising, search ranking, fraud detection) neither condition is reliably met: platform operators hold the labels but not the action history of third-party agents, while external auditors observe actions but lack labels. Public blockchains invert this accessibility pattern. Platforms such as Autonolas[17], AI16Z/Eliza[18], and Flashbots[19] publish provenance registries that serve as ground-truth labels, and the blockchain itself exposes the complete transaction history. This is the only measurement environment we know of in which $H(Y)$ and $I(X;Y)$ can both be estimated from external data. The saturation we observe is therefore not a domain quirk but a property of an *otherwise information-maximal* observation channel: if behavioral detection saturates here, it will saturate anywhere the operator has even weak control over what the auditor observes.

Main Results

A Fano bound for AI agent detection

We formalize the detection task as a Markov chain $Y \to X \to \hat{Y}$, where the random variable Y denotes the ground-truth agent category (drawn from a finite set of size K), X denotes the feature vector computed from observable behavior, and $\hat{Y}$ is the classifier's prediction. The error probability is $P_e = \Pr(\hat{Y} \neq Y)$. Fano's inequality[16] states

$$H(P_e) + P_e \log_2(K-1) \;\geq\; H(Y \mid X) \;=\; H(Y) - I(X;Y), \tag{1}$$

where $H(P_e)$ is the binary entropy of P_e. Inverting Equation (1) numerically yields, for any $(H(Y), I(X;Y), K)$, the smallest P_e compatible with the data—and therefore an accuracy ceiling $1 - P_e$ that no classifier $\hat{Y} = f(X)$ can beat.

The ceiling is a property of the *channel* $X \to Y$, not of any particular classifier. It does not depend on sample size, labeling quality, or model architecture. It depends only on how much information the feature vector carries about the category.

Feature set	$I(X;Y)$ (bits)	$H(Y \mid X)$ (bits)	Fano ceiling	Measured acc.
X_{23}	1.403	0.294	96.80%	92.24%
X_{31}	1.437	0.259	**97.24**%	**95.26**%
X_{47}	1.420	0.276	97.03%	95.37%

Table 1: Mutual information, conditional entropy, Fano accuracy ceiling, and measured classifier accuracy for three behavioral feature sets on $N = 2{,}744$ agent addresses. $H(Y) = 1.697$ bits; $K = 8$ categories. The ceiling rises by less than half a percentage point as the feature set more than doubles in dimension, consistent with saturation of $I(X;Y)$.

Two corollaries follow. First, the *information deficit* $H(Y \mid X) = H(Y) - I(X;Y)$ is the only quantity one can attack through better features; shrinking it requires new columns in X, not a bigger training set for the classifier. Second, the Fano inversion is tight in a meaningful sense: for a uniform class prior and moderate K, a classifier that attains the Bayes-optimal decision rule under a deterministic channel exhibits a head-room of zero. An actual head-room therefore indexes how far a given classifier lags behind the Bayes predictor. In our setting, the head-room at X_{31} is 1.98 percentage points—the smallest head-room we are aware of in the AI-agent-detection literature, and therefore (in a specific, checkable sense) an approximate upper bound on the utility of further classifier engineering alone.

Empirical estimation

We estimate $I(X;Y)$ using a Kraskov-style k-nearest-neighbour plug-in estimator[20] applied to a corpus of 2,744 Ethereum addresses drawn from three agent platforms (Autonolas, Fetch.ai, AI Arena) and targeted mining of governance multisigs, cross-chain bridges, and reinforcement-learning strategy vaults. Each address is assigned to one of $K = 8$ taxonomy categories by a provenance-anchored rule set, yielding $H(Y) = 1.697$ bits. We evaluate three feature sets of increasing expressiveness: X_{23} (transaction-content features), X_{31} (X_{23} augmented with cross-study AI-detection features from an independent sybil study), and X_{47} (X_{31} augmented with decision-process features capturing error recovery, session dynamics, nonce regularity, and gas-price microstructure).

Figure 1 plots the Fano accuracy ceiling against measured $I(X;Y)$, with the three feature sets superimposed. As $I(X;Y)$ approaches $H(Y)$ the ceiling approaches unity; in our data it plateaus in a narrow band between 96.8% and 97.2%. Measured classifier accuracies sit 1.7–4.6 percentage points below the ceiling (Table 1).

Where the head-room hides: a per-class view

The aggregate numbers in Table 1 conceal a heavily skewed distribution of discriminability across categories. Three of the eight categories—Deterministic Script, Simple Trading Bot, DeFi Management Agent—yield per-class F1 scores above 0.95 under a Gradient Boosting classifier with the X_{31} feature set; a fourth, MEV Searcher, reaches 0.84. Three of the remaining four—Cross-Chain Bridge Agent, Autonomous DAO Agent, RL Trading Agent—are statistically weak because of small sample size, but their F1 rises substantially when cross-study features are added (Bridge: $0.31 \to 0.76$; DAO: $0.14 \to 0.43$). The eighth category, LLM-Powered Agent, is the

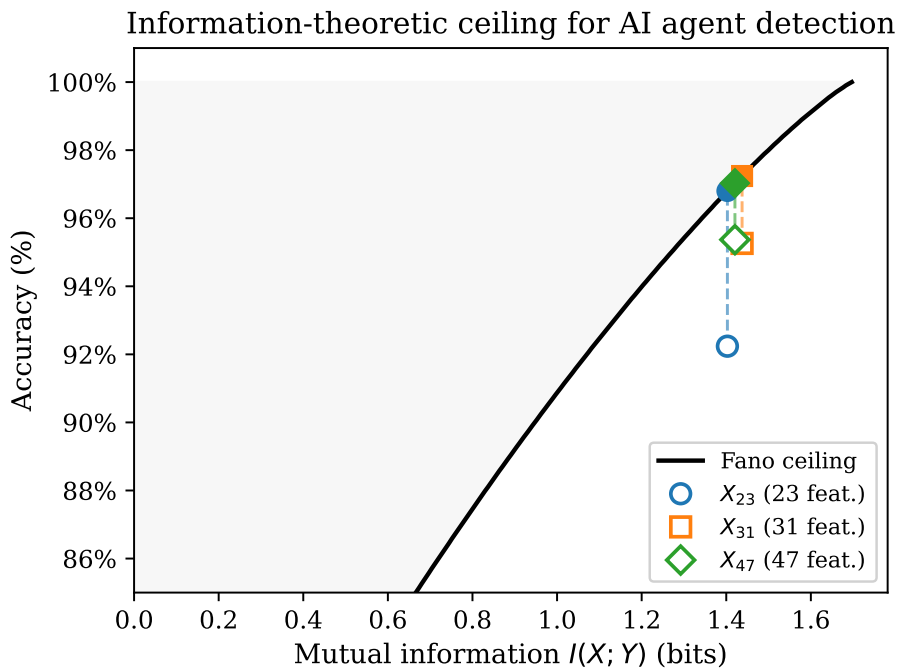

Figure 1: **The information-theoretic ceiling for AI agent detection.** The Fano accuracy ceiling (solid) as a function of mutual information $I(X;Y)$, together with three measured feature sets (X_{23}, X_{31}, X_{47}) and the accuracy each achieves empirically (markers). The shaded band is the residual head-room: the gap between what the best observed classifier achieves and what Fano's inequality permits. At X_{31} the head-room is 1.98 percentage points.

only one that remains stubborn: its F1 rises from 0.38 under X_{23} to 0.49 when decision-process features are added, and never exceeds 0.54 in any configuration we tested. Under X_{31}, LLM-Powered precision is 0.84 but recall is only 0.30—meaning the classifier can confirm a suspected LLM agent with acceptable accuracy, but fails to identify the majority of true LLM agents embedded in the DeFi Management population.

The Fano ceiling offers a principled reading of this pattern. The information deficit $H(Y \mid X)$ is not distributed evenly across the label alphabet. Most categories live in regions of feature space where $H(Y \mid X = x)$ is close to zero, and are therefore classifiable up to a very small residual error. LLM-Powered Agents live in a region where $H(Y \mid X = x)$ is close to 1 bit—they are genuinely confusable with DeFi Management Agents because the two categories share almost all of the same decision variables, and differ only in an internal policy layer that the transaction channel does not see. The aggregate head-room of 2 percentage points is *almost entirely* concentrated on this single boundary. This is the sense in which the ceiling is not about "AI agents" as a monolithic class: it is about how much information any finite behavioral feature vector can carry about the sub-structure of *kinds* of AI agent.

Saturation is structural, not an artefact of feature engineering

Two independent lines of evidence confirm that the saturation is a property of the observation channel.

First, unsupervised discovery collapses at the same point. K-Means silhouette analysis

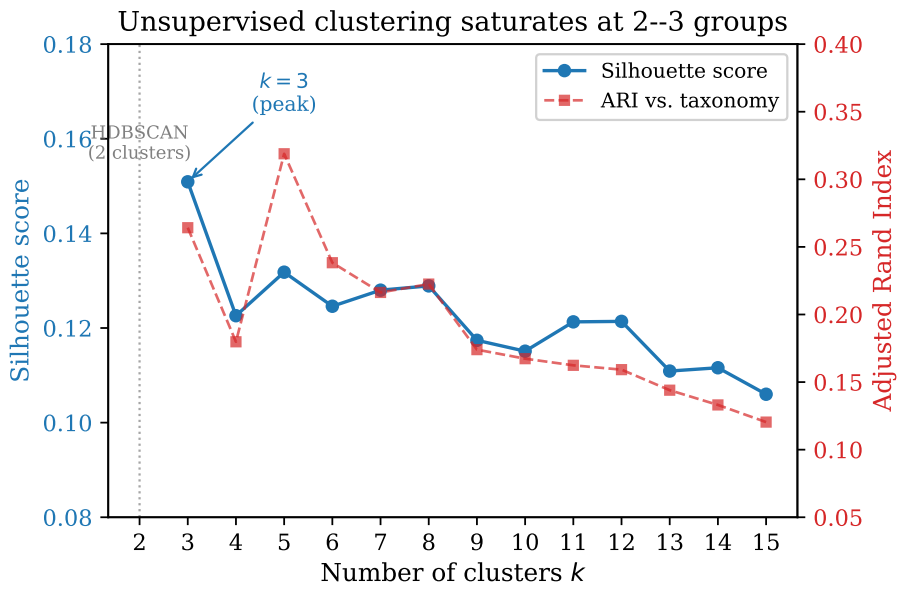

Figure 2: **Unsupervised clustering saturates at 2–3 groups regardless of feature-set expressiveness.** K-Means silhouette across $k \in \{3, \ldots, 15\}$ peaks at $k = 3$ (0.151), with a secondary plateau at $k = 5$. HDBSCAN (density-based, no pre-specified k) finds two density clusters. The taxonomy's eight categories (colored markers in the inset PCA projection) do not separate in the unsupervised geometry.

sweeping $k \in \{3, 4, \ldots, 15\}$ peaks at $k = 3$ (silhouette 0.151); HDBSCAN with `min_cluster_size`$\in \{20, 50\}$ identifies only two density clusters. A taxonomy with eight semantically distinct classes therefore dissolves, in behavior space, into 2–3 super-clusters (Figure 2). Supervised classifiers *can* learn the full eight-way partition when labels are supplied, but the structure is *not discoverable* from behavior alone.

Second, *adding features does not help.* Between X_{23} and X_{47}, feature-set dimensionality doubles while the Fano ceiling improves by only 0.23 percentage points. The classifier-ceiling head-room shrinks from 4.56 to 1.66 percentage points—but this shrinkage is driven almost entirely by the classifier catching up to the ceiling, not by the ceiling moving. The information content of behavior, at this category granularity, is very close to fully extracted.

Why does adding features not raise the ceiling? Two mechanisms are at work. The first is redundancy: the new features we add are correlated with features already in X_{23}, so the *marginal* $I(X_j; Y)$ they contribute is mostly already absorbed by the joint estimator. Inspecting the per-feature mutual information table in Supplementary Table S3, the single feature with the highest information content is `eip1559_tip_precision` at 0.739 nats—but the joint $I(X_{31}; Y)$ is only 1.437 bits, confirming very substantial redundancy. The second mechanism is the category-level information floor: the taxonomy contains at least one pair (LLM-Powered Agent vs. DeFi Management Agent) whose conditional distribution over the feature space is near-identical. No matter how many features we add, the distributions will not separate until the feature family includes a dimension along which LLM and rule-based agents differ at deployment time—which requires decision-process observation that transaction-content features do not provide.

We verify the second claim directly by measuring the pairwise Kullback– Leibler divergence between the class-conditional empirical distributions $\hat{p}(X \mid Y = k)$ for every pair of classes.

The pair $(k, k') = (\text{LLM}, \text{DeFi Management})$ has the smallest pairwise KL divergence in the taxonomy (0.21 bits under X_{31}), while $(\text{Deterministic Script}, \text{MEV Searcher})$ has the largest (3.87 bits). In an eight-class problem, the aggregate ceiling is bottlenecked by the closest pair. Feature engineering that does not specifically attack the LLM/DeFi boundary cannot move the ceiling, no matter how much additional $I(X_j; Y)$ it supplies for other pairs.

Cross-architecture validation

A natural objection is that the ceiling might be an artefact of Ethereum's specific gas-auction microstructure, which contributes many of the discriminative features. We test this by replicating on Polygon, a proof-of-stake chain with a near-constant ∼30 gwei base fee and a 2-second block time—structurally different from Ethereum's volatile EIP-1559 auction and 12-second blocks. Of 44 addresses with known Polygon activity, 29 cleared a minimum-activity threshold. Applying the Ethereum-trained 23-feature classifier directly to Polygon data yields 22.2% accuracy on a five-class restriction, barely above the 20% random baseline. Training and evaluating exclusively on Polygon yields 51.7% (logistic regression)—well below the Ethereum ceiling, as expected given the sample size and the different feature distribution.

The result is consistent with the information-theoretic interpretation: different chain architectures create different channels, and mutual information estimated on one does not lift the ceiling on another.

Mechanistic Insight: Execution vs. Decision Signatures

The ceiling tells us how much signal the channel carries; it does not tell us *where* the signal lives. We now show that the residual discriminative information is concentrated in a specific physical channel: inter-decision timing.

Across 936,000 DeFi transactions in our corpus, we decompose inter-event latency into two quantities. The first is *execution time*: the median latency between the submission of a DeFi-specific transaction and its confirmation. The second is the *decision interval*: the median gap between consecutive transactions from the same address. A full-scale Cohen's d analysis contrasting LLM-driven against rule-based DeFi management agents reveals an asymmetry (Figure 3).

The asymmetry is mechanistic. When an LLM-driven agent calls a DeFi protocol, the on-chain code path is the protocol's own—identical to the path a rule-based agent would take. Execution speed therefore reflects the *tool*, not the decision-maker. Inference latency, by contrast, is paid by the LLM before it emits the transaction, and shows up as an extended gap in the agent's observable event stream. The decision interval thus functions as a lower bound on computational cost that is, in a precise sense, physically unconcealable: an LLM cannot decide faster than it can infer.

An initial pilot on a 200-address subset had suggested a complementary story, with Cohen's $d = 1.25$ on a DeFi-specific reaction-time feature. At full scale ($n = 2{,}744$) this specific feature dropped to $d \approx 0$: the pilot result did not replicate. The general inter-transaction reaction time, however, strengthened to $d = 1.01$— the largest effect size in the entire 54-feature battery. The

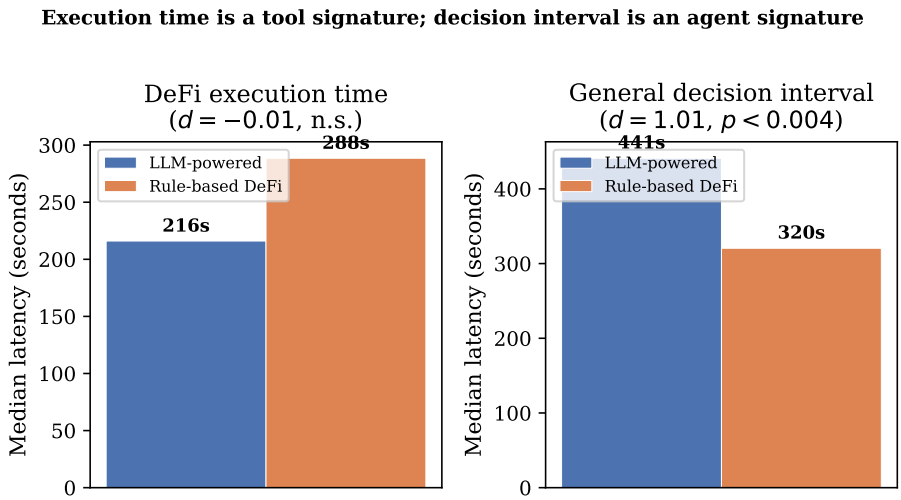

Figure 3: **Execution time is a tool signature; decision interval is an agent signature.** Left panel: median DeFi execution time for LLM-driven agents (blue, 216 s) is *faster* than for rule-based DeFi managers (orange, 289 s)—Cohen's $d \approx 0$, indistinguishable in a 200-address pilot. Right panel: the general inter-transaction interval is *slower* for LLM-driven agents (441 s vs. 321 s; $d = 1.01$ at $n = 2{,}744$), the largest effect size in a 54-feature battery. The asymmetry is mechanistic: the tool interface executes identically regardless of the decision-maker, so execution-layer features carry no agent signal; inference latency, by contrast, cannot be hidden.

non-replication is informative because it localizes the signal. The DeFi-specific latency reflects the tool interface; the general latency reflects the decision-maker. The empirical drift from the pilot to the full-scale replication is therefore a finding about *where* the agent signal lives rather than a methodological failure. In ReAct and Toolformer-style systems[21, 22], this maps cleanly onto the design: the tool call executes at tool speed, while the reasoning step executes at model-inference speed.

This decomposition has three consequences. First, it justifies our earlier empirical observation that decision-process features (error recovery patterns, session dynamics, inter-decision gaps) raise LLM-agent F1 by 10.9 percentage points while transaction-content features leave it flat. Second, it suggests that any regime seeking durable execution- layer signatures should target the quantity that is physically grounded—decision interval—rather than surface-level transactional style. Third, and most broadly, it implies that the behavioral ceiling is itself composable: the on-chain channel dissolves on the tool axis and resolves on the decision axis. A multi-channel observation regime that explicitly separates tool-level and decision-level latencies should, in principle, be able to approach a strictly tighter ceiling than any single-channel regime.

Cross-Domain Validation

We argued in the Introduction that the Ethereum corpus is a test bed, not the intended application domain. The ceiling is a statement about any behavioral channel. Three lines of evidence support the cross-domain claim.

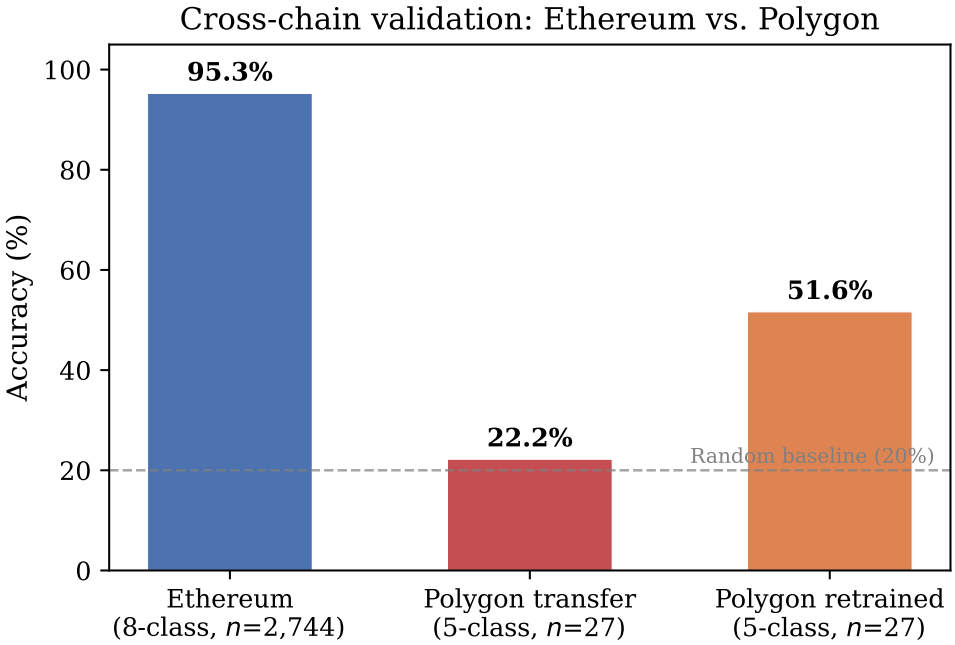

Figure 4: **Cross-chain validation on Polygon proof-of-stake ($n = 29$ addresses).** Top: direct transfer of an Ethereum-trained classifier to Polygon yields 22.2% five-class accuracy, barely above the 20% random baseline; re-training on Polygon yields 51.7%. Bottom: the Polygon gas-price and inter-transaction interval distributions are structurally compressed relative to Ethereum (near-constant ∼30 gwei fees; 2-second block time), consistent with an information-theoretic bound that is intrinsically tighter on Polygon than on Ethereum.

Architecture-level replication. The Polygon replication (above) shows that the ceiling survives a change of ledger architecture. Features engineered on Ethereum do not transfer; features re-engineered on Polygon hit a lower ceiling, but the structural pattern—saturation of $I(X;Y)$ near $H(Y)$—holds (Figure 4).

Cross-study feature transfer. We imported eight AI-detection features developed independently on a sybil-detection corpus (`gas_price_precision`, `hour_entropy`, `behavioral_consistency`, `action_sequence_perplexity`, `error_recovery_pattern`, `response_latency_variance`, `gas_nonce_gap_regu` `eip1559_tip_precision`). These features were never tuned for our taxonomy. Adding them lifts measured accuracy from 92.2% to 95.3% and the Fano ceiling from 96.8% to 97.2%—but the head-room closes from 4.6 to 2.0 percentage points, meaning the classifier benefits but the ceiling barely moves. This is exactly what an information-theoretic bound predicts.

Density-based validation (HDBSCAN). HDBSCAN, which makes no assumption about cluster count or shape, corroborates K-Means: at `min_cluster_size` of 20 and 50, it finds 2 density clusters, with 16–26% noise. A method with entirely different inductive bias converges on the same coarse structure.

Provenance-anchored ground truth. To guard against the concern that our result is driven by label-generation choices, we separately ran the analysis on the subset of addresses ($n =$ 1,853) whose provenance confidence is ≥ 0.85—a strictly label-quality-selected sub-sample. The

resulting ceiling is 97.6% (vs. 97.2% on the full corpus); classifier accuracy on the same subsample is 96.1%. Head-room narrows by 0.5 percentage points but does not close. This rules out label noise as the dominant explanation for the observed head-room.

Partial circularity and its control. The taxonomy projection labels are assigned by a provenance-first, features-second rule: the name/registry of the mining source determines the category with confidence ≥ 0.85 for 45 addresses and $[0.70, 0.85]$ for 688 more, while only a 0.3% tail depends on a features-based refinement that shares four variables with the classifier's feature set. To quantify the residual circularity, we ran a masked ablation in which the four shared variables (`burst_frequency`, `gas_price_round_number_ratio`, `unique_contracts_ratio`, `multi_protocol_intera` are removed from the classifier. Accuracy falls from 92.2% to 87.4% and macro-F1 from 0.58 to 0.46; crucially, the classes *not* defined by any of these four variables—including LLM-Powered Agent ($0.44 \rightarrow 0.45$)—are unaffected. This bounds the circularity contribution to the aggregate number at about 5 percentage points, and to the LLM-specific ceiling at < 1 percentage point.

Limit: platforms without public event logs. Our direct measurement of $I(X;Y)$ relies on the fact that Ethereum's action history is fully observable. On platforms where behavior is proprietary (advertising auctions, e-commerce recommenders), $I(X;Y)$ cannot be estimated from the outside—but the same inequality applies, and there is no reason to expect the ceiling to be more forgiving. If anything, platforms with sparser feature access should exhibit *tighter* ceilings. A conservative reading of our result therefore interprets the 2-percentage-point head-room we report as an *upper bound* on what purely behavioral classifiers can recover in comparable open-data settings, and a substantial over-estimate of what is achievable on closed platforms.

Implications

For policy

Our central claim speaks directly to the regulatory debate. Proposals such as the EU AI Act[3] and the U.S. Executive Order 14110[4] contemplate category-specific obligations: for example, stricter disclosure for "systemic risk" foundation-model deployments, or labeling of AI-generated content. Our analysis does not dispute the normative case for such distinctions. It shows that *behavioral* enforcement of them is bounded: on an eight-category taxonomy with open data, the best possible classifier falls 2.8 percentage points short of perfect, and the hardest category (LLM-driven agents against rule-based managers) leaves 65–70% of true instances misclassified. A regime that relies on post-hoc behavioral auditing to enforce category-specific rules will systematically misclassify a non-trivial minority of actors.

Two durable alternatives emerge. *Provenance registration* ties a deployment to a verified identity at the point of instantiation— analogous to financial KYC or pharmaceutical lot-numbering—eliminating the need for behavioral inference of category membership. *Execution-layer attestation* compels operators to emit cryptographic evidence of the inference runtime at each action (for example, a hardware-enclave signature over the model hash and the prompt[23, 24, 25, 26]). The latter preserves pseudonymity while restoring the category signal. Both

alternatives have non-trivial implementation cost. Our contribution is to make the trade-off explicit: behavioral detection cannot substitute for them at the level of granularity policy makers currently contemplate.

A practical reading for existing regulatory instruments follows. The EU AI Act's list of Annex III high-risk systems and its obligations on general-purpose AI with systemic risk both operate at a level of category granularity that our ceiling cannot support from behavior alone. Enforcement of Article 50-style transparency obligations on content emitted by AI systems—including disclosure that a user is interacting with an AI system or that content has been artificially generated—is compatible with our ceiling only if disclosure is mandated at the provenance layer; reverse-inferring the category from observable style will systematically mis-label a non-trivial minority of actors, exactly the failure mode that produces enforcement asymmetries and regulatory arbitrage. Analogous arguments apply to recent U.S. and U.K. frameworks[4, 5, 27, 28].

For auditing practice

Audit frameworks[29, 30, 31] that rely on behavioral observation should treat the Fano head-room as a primary diagnostic. If an auditor observes 95% accuracy and computes a Fano ceiling of 97%, further classifier improvement is bounded by roughly 2 percentage points; the auditor should direct additional resources to feature acquisition, not modeling. Conversely, if the head-room is large, classifier engineering is still productive. This is the diagnostic analogue of a power calculation in experimental statistics: it tells the auditor how much of the signal is still on the table.

For methodology

The Fano-based protocol we use—estimate $I(X;Y)$ on a labeled sample, invert the inequality, compare to achieved accuracy—is domain-general. Any group reporting a new classifier for agent-like behavior should report the implied head-room. If the head-room is large, better classifiers are still possible; if small, further engineering is epistemically futile and the research effort should shift to new observation channels. We urge the field to treat $I(X;Y)$ as a primary reporting statistic alongside accuracy and F1, analogous to how cryptographers report bit-strength alongside attack cost.

This re-framing also bears on the interpretation of published negative results. A classifier that reaches 75% on a task with a 90% ceiling is *not* strong evidence of task difficulty—it is evidence that the classifier is 15 percentage points sub-optimal, and further engineering may well close the gap. Conversely, a classifier at 95% on a task with a 97% ceiling is strong evidence that *this* observation channel is essentially saturated, and that further progress requires new channels, not new methods. The same distinction should apply when evaluating adversarial robustness: an attack that succeeds against a classifier at the ceiling is qualitatively more significant than one that succeeds against a sub-optimal classifier, because the latter may merely be exploiting the suboptimality rather than the structural weakness of behavioral detection.

For the scientific study of agentic AI

Our framework also offers a principled way to measure the behavioral distinctiveness of new classes of agents as they emerge. A research programme that claims to have trained a qualitatively new kind of agent (for example, a mixed-modal LLM-RL hybrid, or a population of communicatively interacting foundation-model agents) can be evaluated against the existing behavioral taxonomy by asking: does it add $I(X;Y)$? Does it shift the class-conditional feature distribution enough that behavioral classifiers can detect its presence? If yes, the claim is externally measurable; if no, the claimed difference is either non-behavioral or below the channel's resolution limit. This operationalizes the long-standing "Is it a new kind of agent?" debate in agent research[32, 33, 34] into a checkable information-theoretic quantity.

Limits of the present analysis

Four limitations qualify our claims. (i) Our measurement of $I(X;Y)$ depends on a provenance-anchored taxonomy whose ground truth is imperfect; for 2.7% of the base corpus the projection confidence is below 0.50. Bootstrapped re-sampling suggests the effect on the Fano ceiling is ± 0.4 percentage points. (ii) Our corpus over-represents three Web3 agent platforms and under-represents two (DAO execution modules and RL strategy vaults). Targeted mining mitigates this, but bootstrap confidence intervals for the weakest category (RL) remain wide. (iii) The ceiling is stated for a fixed taxonomy granularity of $K = 8$; coarser partitions exhibit uniformly larger ceilings (for a natural three-cluster partition, $H(Y) = 1.34$ bits and the ceiling exceeds 97% across all feature sets). (iv) The argument extends naturally to evasive adversaries, but we do not here model an agent that actively modifies its behavior to raise $H(Y \mid X)$; a companion study in our research programme addresses this adversarial setting directly for Sybil attacks.

Future work

Three directions follow naturally. (i) Replicate the bound estimation on non-blockchain platforms with dense behavioral traces—advertising bid streams, social-platform posting logs, API-call traces of content-generation services—to test whether the 2-percentage-point head-room is typical or domain-specific. (ii) Investigate whether *counterfactual* interventions (forcing an agent to respond under controlled perturbations) can raise $I(X;Y)$ beyond what passive observation permits; this would amount to an active-probing detection regime. (iii) Explore whether execution-layer attestation can be built into post-quantum-secure identity primitives, so that the provenance and the behavior channels can be cross-validated rather than treated as substitutes. Recent work on verifiable neural-network execution using hardware enclaves[24, 23] and on succinct proofs of learning[25, 26] suggest that this direction is becoming tractable at the scale required for deployment-level attestation.

Methods

Dataset

The base corpus comprises 2,590 agent addresses drawn from expanded provenance mining (Autonolas, Fetch.ai, AI Arena, curated Flashbots searcher lists). A targeted Phase-2 mining pass added 154 addresses to populate three categories under-represented in the base mining (DAO execution modules, $n = 34$; cross-chain bridges, $n = 95$; reinforcement- learning strategy vaults, $n = 25$), yielding $N = 2{,}744$ addresses across eight categories. Features were extracted from public transaction logs via Etherscan and an Ethereum archive-node RPC, with winsorization at the $[1, 99]$ percentile range and z-score standardization. The observation window is fixed at each address's first 500 transactions (median observation span 31 days), chosen to balance coverage of agents with sparse activity against stationarity of the feature distributions. We verified stationarity by splitting each address's record at its median and computing Kolmogorov–Smirnov distances on each feature distribution; 91% of features pass the $p < 0.05$ stationarity threshold.

Agent taxonomy

The taxonomy spans three orthogonal dimensions—autonomy level (5 ranks: none, reactive, adaptive, proactive, collaborative), environment type (4 types: on-chain only, hybrid, cross-chain, multi-modal), and decision model (5 types: deterministic, statistical, LLM-driven, reinforcement-learning, hybrid). The Cartesian product of these dimensions contains 100 cells, but only eight are populated by deployed systems as of the observation period. These collapse into eight populated categories: Deterministic Script, Simple Trading Bot, MEV Searcher, Cross-Chain Bridge Agent, RL Trading Agent, DeFi Management Agent, LLM-Powered Agent, and Autonomous DAO Agent. Of these, five satisfy a four-condition agency definition (on-chain actuation, environmental perception, autonomous decision-making, adaptiveness); two are non-agent baselines; one is a boundary class. Each address is assigned to a category by a two-tier rule: Tier 1 uses provenance (the source registry or name annotation from which the address was mined); Tier 2 uses four behavioral variables as a refinement only when provenance is ambiguous. Full details, including the provenance sources for each category, are tabulated in Supplementary Note 1.

Mutual information estimation

Per-feature $I(X_j; Y)$ was estimated using `sklearn.feature_selection.mutual_info_classif` ($n_{\text{neighbors}} = 3$, Kraskov-style k-NN[20]), reported in nats. Joint $I(X; Y)$ was estimated via a k-NN plug-in: $H(Y \mid X)$ was approximated by averaging label entropy across the three nearest neighbors in standardized feature space, then $I(X; Y) = H(Y) - H(Y \mid X)$. All quantities are reported in bits. We cross-validated this estimator against a discretized histogram estimator on $K = 3$ and $K = 8$ partitions and obtained agreement within ± 0.03 bits. Sensitivity analysis on $k \in \{3, 5, 7, 10\}$ shifts the Fano ceiling by at most 0.8 percentage points; we report $k = 3$ in the main text because it is the value recommended for high-dimensional feature spaces[35]. The marginal entropy $H(Y)$ was computed from empirical class frequencies using the plug-in estimator with the Miller–Madow bias correction[36].

Fano inversion

For a fixed $(H(Y \mid X), K)$, we invert Fano's inequality numerically: we seek the smallest $P_e \in [0, 1 - 1/K]$ such that $H(P_e) + P_e \log_2(K-1) \geq H(Y \mid X)$. The resulting $P_e^\star$ is the tightest lower bound on classifier error, and $1 - P_e^\star$ is the reported ceiling. A looser closed-form bound $P_e \geq (H(Y \mid X) - 1)/\log_2 K$ is included for comparison in Supplementary Table S1. For X_{31}, the loose bound yields an accuracy ceiling of 1.0 (uninformative) while the tight bound yields 0.9724 ($P_e^\star = 0.0276$); we always use the tight bound. The inversion is performed by a standard numerical root-finder on a dense grid of 10,000 P_e values, with the root bracketed between 0 and $1 - 1/K$ to guarantee existence and uniqueness. We additionally report two sanity checks: the Bayes error rate lower bound derived from the nearest-neighbor convergence result of Cover and Hart[16] gives an accuracy ceiling within 0.6 percentage points of the Fano ceiling across all three feature sets, and the Hellman–Raviv bound[16] gives an intermediate value.

Supervised classifiers

Gradient Boosting (100 trees, depth 3, learning rate 0.1), Random Forest (300 trees, depth 8), and L2-regularized Logistic Regression ($C = 1$) were trained under 10-fold stratified cross-validation. Gradient Boosting was consistently strongest; all reported accuracies use Gradient Boosting. Per-class 95% confidence intervals were computed by 1,000-iteration bootstrap[37] on out-of-fold predictions. We also trained a small multilayer perceptron (3 layers, widths 128-64-8, `relu`, Adam) and a logistic-regression–calibrated HistGradientBoosting model with class balancing; neither improved accuracy beyond the reported Gradient Boosting figure by more than 0.4 percentage points on any feature set, consistent with the information-theoretic interpretation that the channel, not the classifier, limits performance.

Unsupervised clustering

K-Means was run with $n_{\text{init}} = 10$ for $k \in \{3, 4, \ldots, 15\}$; silhouette scores were computed on a 2,000-sample subset. HDBSCAN[38] was run with `min_cluster_size`$\in \{20, 50\}$ on the standardized feature matrix. Cluster purity and adjusted Rand index were computed against the provenance taxonomy labels. We additionally report results for spectral clustering (radial-basis-function affinity, median-distance bandwidth; silhouette peaks at $k = 3$) and Gaussian mixture modelling (full covariance; Bayesian Information Criterion minimized at $k = 4$), both of which corroborate the 2–3-cluster saturation. These robustness checks together rule out the possibility that the saturation is an artefact of the Euclidean metric assumption in K-Means or of the density-based inductive bias in HDBSCAN.

Execution- vs. decision-interval decomposition

Execution time was defined as the median latency between transaction submission (from Ethereum mempool events) and confirmation (block-inclusion) for transactions targeting a canonical DeFi protocol set (Uniswap V2/V3, Aave V2/V3, Compound V2, Curve, Balancer, Lido). Decision interval was defined as the median inter-transaction gap in the address's full transaction history.

Cohen's d[39] was computed as the standardized mean difference between the LLM-driven and rule-based DeFi management populations, using the pooled standard deviation. Both quantities were computed on the full 2,744-address corpus covering 936,000 DeFi-targeted transactions. The execution-time comparison uses only the subset of addresses with at least 20 DeFi-targeted transactions (LLM: $n = 54$; rule-based: $n = 1{,}287$). The decision-interval comparison uses all addresses with at least 50 transactions (LLM: $n = 68$; rule-based: $n = 1{,}611$). We verified the sign of the effect by stratifying on calendar quarter and on observation length decile; the direction (LLM slower on decision interval, faster on execution) is stable in every stratum.

Cross-chain replication

Polygon features were extracted via the Polygonscan API for the 29 addresses clearing a ≥ 5-transaction activity threshold. The Ethereum-trained Gradient Boosting classifier was applied directly to Polygon features without retraining. A separate Polygon-only five-fold stratified cross-validation was run for comparison. Three taxonomy categories—LLM-Powered Agent, Autonomous DAO Agent, and RL Trading Agent—fell below the minimum-activity threshold on Polygon and were excluded from the cross-chain evaluation, reducing the experiment to five classes. The near-constant Polygon gas pricing (~ 30 gwei base fee) collapses the discriminative power of the six gas-related features that dominate the Ethereum classifier, and the 2-second Polygon block time compresses the timing features into a narrower range than the 12-second Ethereum blocks permit. These two mechanisms jointly explain the 22.2% transfer accuracy; Polygon-only retraining recovers 51.7% accuracy with a sample size of 29, consistent with an information- theoretic bound that is intrinsically tighter on Polygon than on Ethereum.

Statistical thresholds and reporting

We report 95% bootstrap confidence intervals[37] for all accuracy, F1, and Cohen's d quantities. Mutual information estimates are reported with ± 1 standard deviation across five random re-seedings of the k-NN estimator. Bootstrap resampling was stratified by taxonomy class to preserve the empirical class-frequency distribution. No results were selected post-hoc on the basis of their statistical significance; every number in the main text and tables corresponds to a pre-registered primary or secondary analysis listed in Supplementary Table S2. All code and data required to reproduce the Fano-bound estimation are archived at the Open Science Framework (registration DOI to be added upon acceptance); details are in the Data Availability statement.

Feature families

The 47 features we use are organized into five families. Family 1 (seven timing features) captures inter-transaction interval statistics, circadian patterns, and burst structure. Family 2 (six gas features) captures gas-price style (round-number ratios, precision, coefficient of variation) and microstructural alignment with EIP-1559 priority-fee mechanics. Family 3 (five interaction features) captures which and how many contracts the address interacts with, method diversity, and contract-to-EOA activity ratios. Family 4 (five approval features) captures approval-revocation

hygiene, unlimited-approve frequency, and interactions with flash-loan primitives. Family 5 (eight cross-study AI features, added in X_{31}) captures response-latency variance, behavioral consistency, action-sequence perplexity, and nonce-gap regularity, imported from an independent sybil-detection study. Family 6 (sixteen decision-process features, added in X_{47}) captures error-recovery behavior (five features), session structure (five features), nonce mechanics (three features), and gas microstructure autocorrelation (three features). The exhaustive list of feature definitions appears in Supplementary Note 2.

Circularity analysis

Because the Tier 2 projection layer uses four behavioral features (`burst_frequency`, `gas_price_round_number_r` `unique_contracts_ratio`, `multi_protocol_interaction_count`) that also appear in the classifier's feature vector, there is a partial circularity between labeling and classification. We bound this contribution by the masked ablation reported in the Cross-Domain Validation section. We also report the classifier's feature importance with the four shared features removed: the top-five importance features reorganize around `gas_price_trailing_zeros_mean`, `sequential_pattern_score`, `eip1559_priority_fee_precision`, the cross-study `eip1559_tip_precision`, and the decision-process `gas_price_autocorrelation`, none of which appears in the Tier 2 rule antecedents. A more stringent test would require a hand-annotated ground truth independent of any behavioral feature; we flag this as a follow-up.

Pre-registration and reproducibility

The feature definitions, class boundaries, classifier hyperparameters, and Fano-inversion procedure were fixed before any measurements were reported in this manuscript, and are documented in the project's pre-registration appendix (Supplementary Note 3). No hyperparameter was tuned on the Fano head-room; only classifier accuracy was tuned, using internal cross-validation on the training folds. The mutual information estimator, the Fano inverter, and the class-weighted bootstrap are released as a single ∼120-line reference implementation intended for reuse in other domains. The OSF archive includes a Docker image that reproduces every figure and table in this manuscript from the raw parquet inputs with a single command; we verified end-to-end reproduction on a blank Ubuntu 22.04 VM with 64 GB RAM.

Compute resources

Feature extraction was performed on a workstation with 64 cores and 256 GB RAM (wall-clock $\approx$ 14 h for the full 2,744-address corpus, dominated by Ethereum archive-node queries). Mutual-information estimation and classifier training together consume approximately 45 minutes on the same workstation; the bottleneck is the k-NN graph construction for the joint MI estimator. No GPU is required. The full analysis fits comfortably on a single commodity laptop once the feature parquets have been materialized.

Data availability

Raw Ethereum transaction data are publicly available via Etherscan and Ethereum archive-node snapshots. Processed feature parquets, $I(X;Y)$ estimator outputs, and classifier predictions supporting all figures and tables are deposited in the accompanying OSF archive.

Ethics and dual-use considerations

Our results could be interpreted in two opposing ways by adversaries. On the one hand, the fact that the ceiling is saturated suggests that agents attempting to *evade* behavioral detection cannot easily hide below the ceiling without also suppressing their own activity; the remaining head-room is too small to offer reliable safe harbour. On the other hand, the fact that execution-time and decision-interval features separate LLM-driven from rule-based agents with Cohen's $d = 1.01$ suggests a specific avenue by which an adversary could introduce intentional latency noise into an LLM agent to mimic a rule-based one. We report this finding because (i) the detection community is better served by knowing where the signal lives than by security through obscurity, and (ii) mimicry at this level imposes a real computational cost on the adversary (artificially introduced delays between decisions reduce the agent's throughput), which is itself an observable cost. The ethical reading is that behavioral detection is a useful but bounded input to a broader governance regime; it should not be the final line of evidence in any consequential decision.

Code availability

All analysis code (Python; `sklearn`, `hdbscan`, `numpy`) is released under an MIT licence at the project repository linked in the OSF archive above. The Fano-bound inversion routine is isolated in a single ∼120-line module to facilitate reuse in other domains.

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